CALL FOR PARTICIPATION

APEX-AI: Approximate, Low-Precision, Sparsity, and Efficient AI Execution from Training to Deployment

APEX-AI
APPROXIMATE • EFFICIENT • DEPLOYABLE

• 19 Jan. 2027 • Glasgow, UK • Co-located with HiPEAC 2027

ABOUT THE WORKSHOP

APEX-AI explores approximate and efficiency-oriented techniques for next-generation AI systems. As AI models continue to grow in size and complexity, improving efficiency increasingly requires coordinated advances across algorithms, software, and hardware. The workshop focuses on techniques such as low- and adaptive-precision computation, sparsity and conditional execution, speculative and approximate inference, compressed memory and data movement, and accelerator–compiler–runtime co-design. APEX-AI aims to bring together researchers from architecture, systems, machine learning, and design automation to discuss emerging approaches for improving the performance, energy efficiency, and scalability of AI workloads while managing accuracy and quality trade-offs.

IMPORTANT DATES

Contribution deadline (Extended)	**3 Dec. 2026 (AoE)**
Notification	**8 Dec. 2026**
Workshop	**19 Jan. 2027**

WEBSITE & SUBMISSION

WORKSHOP WEBSITE

SUBMIT AN ABSTRACT

CALL FOR PARTICIPATION

APEX-AI is a curated technical workshop with invited talks from academia and industry, complemented by a small number of contributed presentations selected through an open call. We invite one-page abstracts submissions describing new research, work in progress, preliminary results, experience reports, emerging challenges, or technical perspectives related to approximate and efficient AI systems. No submission is required to attend; participants should register for HiPEAC 2027.

TOPICS OF INTEREST

- **Low-Precision & Approximate AI** — FP8/FP4/INT4, adaptive precision, approximate arithmetic
- **Sparsity & Conditional Execution** — structured sparsity, MoE, dynamic routing, token/layer skipping
- **Memory & Data Movement** — KV cache, compression, sparse access, HBM, interconnects
- **Efficient Large-Model Execution** — speculative decoding, long context, distributed/disaggregated inference
- **Accelerator–Compiler–Runtime Co-Design** — heterogeneous acceleration, scheduling, cross-layer optimization

SUBMISSION GUIDELINES

- Submit a one-page PDF in English with the presentation title, presenter name(s), affiliation(s), and contact email.
- Briefly describe the proposed talk, key technical message, relevance to APEX-AI, and expected discussion value.
- Proposals may cover new work, work in progress, preliminary results, experience reports, or previously published/presented work.
- No formal proceedings are planned; proposals are intended to be non-archival.
- Selected contributors will give a 15–20 minute technical presentation followed by discussion.
- At least one presenter must register for HiPEAC 2027 and attend the workshop.

ORGANIZERS

Chair: Henk Corporaal (Eindhoven University of Technology) **Co-Chair:** Stefano Corda (Huawei Europe)

Organizers: Dimitrios Soudris (National Technical University of Athens), Akash Kumar (Ruhr University Bochum), Philipp Petersen (University of Vienna), Sajjad Tamimi (Independent Technologist), Zheng Li (Huawei Europe)



AI generated images (OpenAI)